In the end, though, I felt these issues were important enough to bring to your attention. Hopefully, you should now be aware of the main problems and prepared to face them.

Let's review what we covered in this chapter:

1. We explored an issue where Rails' tmp/pids/server.pid file doesn't always get removed upon terminating the containers.
2. We learned about *entrypoints*, which are prepended to the command run on starting a new container.
3. We utilized an entrypoint to create a wrapper script that deletes the tmp/pids/server.pid on starting a container, working around the issue.
4. We discussed an issue where Compose intermittently aborts instead of terminating containers gracefully, deciding that the best approach to avoid it may be running Compose in detached mode (-d.)

OK, enough about annoyances. It's time to think about the positives.

Closing Thoughts on Docker in Development

As we draw to the close of this section on development, let's take a moment to reflect on what we've achieved. At first glance, it's easy to mistakenly think we've gone through a fair amount of effort only to arrive back where we started—with a standard, working Rails app.

In fact, though, we've achieved some major benefits:

- Our Dockerfile and docker-compose.yml file give us a declarative description of our entire application—with all its required parts, such the database—helping to give a clear picture of what makes up the application.
- We can spin up the entire application with a single command—even with nothing previously installed. Docker downloads and installs what we need.
- We've eliminated the need to manually install our app's main software dependencies on our local machine. No more fiddling with getting Redis, Postgres, or even Ruby installed and running on compatible versions across the whole team. Docker is taking care of all of this for us.
- That last point is a big one. It also means that our app can run on any machine with Docker installed. It gives us freedom and portability.
- Upgrading parts of our application is as simple as updating the version number of the image we refer to in our Compose file. It's a breeze to see how our app works on a newer version of Ruby, for example.